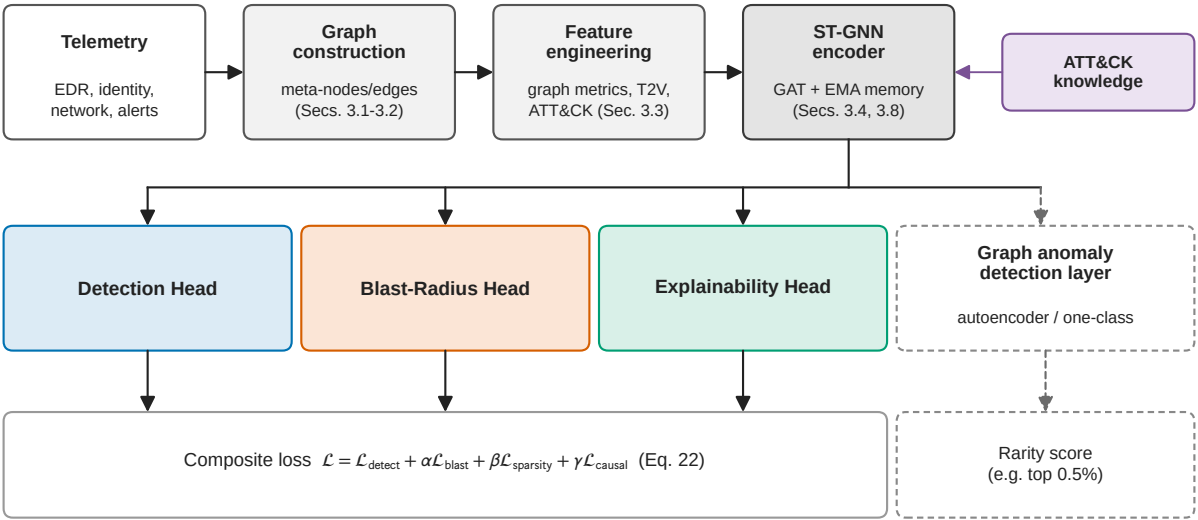
Telemetry
EDR, identity, network, alerts
Graph construction
meta-nodes/edges (Secs. 3.1-3.2)
Feature engineering
graph metrics, T2V, ATT&CK (Sec. 3.3)
ST-GNN encoder
GAT + EMA memory (Secs. 3.4, 3.8)
ATT&CK knowledge
Detection Head
Blast-Radius Head
Explainability Head
Graph anomaly detection layer
autoencoder / one-class
Composite loss $\mathcal{L} = \mathcal{L}_{\text{detect}} + \alpha\mathcal{L}_{\text{blast}} + \beta\mathcal{L}_{\text{sparsity}} + \gamma\mathcal{L}_{\text{causal}}$ (Eq. 22)
Rarity score
(e.g. top 0.5%)